Fishing for Sustainability

Spring 2025

Learning Objectives

By the end of this activity, you should be able to:

1. Explain the concepts of carrying **capacity and maximum sustainable yield (MSY)**.
2. Compare different management strategies—Open Access, Fixed Quota, and Fixed Effort—and their outcomes for resource sustainability.
3. Analyze how cooperative vs. individual incentives influence resource use decisions (the *Tragedy of the Commons*).
4. Reflect on real-world implications of fisheries management strategies and how they might be adapted to other renewable resources.

Background & Context

In natural resource management, fisheries provide a classic example of how unregulated harvesting can lead to **resource depletion—** sometimes rapidly and unexpectedly. This phenomenon is known as the **Tragedy of the Commons**, where individuals, acting in their own self-interest, collectively over-exploit a shared resource.

In fisheries, managers often propose strategies to avoid over harvesting. **Fixed Quota** (limiting total catch) and **Fixed Effort** (limiting how many fishers or how long they can fish) are two common ways to stay near a targeted **Maximum Sustainable Yield (MSY)**. However, each strategy has benefits and drawbacks. This game will illustrate those concepts firsthand.

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Materials & Setup

Groups

- Form groups of 4–6 students.

Reservoir (Bowl)

- Start with 30 fish (e.g., small candies, beans, or tokens).

Boat (Cup)

- Each student has their own "boat" to hold catches.

Notecards or scratch paper

- For privately recording fishing decisions.

Important Biological Rule After each round, any fish left in the reservoir double in number, up to a maximum of 30.[1]

[1] This represents a simplified model of population growth under a carrying capacity.

Data Collection & Analysis

Throughout Rounds 1–3, record data in a simple table:

Round	Strategy	Total Catch	Fish Remaining After Harvest	Fish After Reproduction
1.	Open Access			
2.	Fixed Quota			
3.	Fixed Effort			

Round 1: Open-Access Fishing

Scenario

- Each of you is an independent fisher aiming to catch as many fish as possible.
- **No communication** or coordination is allowed.
- After harvesting, the remaining fish reproduce: if `( x )` fish remain, the population becomes `(\min(2x, 30) )`.

Instructions

1. **Decide privately** how many fish you will catch (up to 6).
2. **Simultaneously reveal** your decisions.
3. **Distribute the fish** to each person's "boat."[2]
4. **Reproduction:**

a. Count the remaining fish.
b. Double that number, up to a max of 30.

[2] If the total catch exceeds what's available in the reservoir, fish are taken first-come, first-served until the reservoir is empty.

Round 1 Reflection Prompt: Discuss with your group and respond on a

c. Record the data: fish caught (per person + total) and fish remaining.

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Before moving on, think about the **maximum number of fish** the population can yield each round without declining over time. This is the **Maximum Sustainable Yield (MSY)**. Because our population doubles each round (with a cap at 30), the MSY will be near the midpoint of carrying capacity. If you remove too many, you risk dropping below the replenishment threshold; if you remove too few, you might not maximize yields.

Round 2: Fixed-Quota Harvesting

Scenario

- You will set a **total catch quota** for each round. (A good starting point is ~15 fish total per round.)
- **You may openly discuss** what the quota should be, but **you cannot reveal your personal take** once fishing begins.
- Your individual goal remains to maximize your own fish caught over multiple rounds.

Instructions

1. **Group Quota Decision (1–2 minutes):**

- Decide on a single group quota (e.g., "We'll catch exactly 15 fish total this round.")

2. **Private Decision:**

- Each person secretly records how many fish they will catch (up to 6).

3. **Reveal & Check Quota:**

- If the total catch exceeds the quota, the fishery collapses immediately: *No fish survive to reproduce this round.*
- If total catch = quota, distribute the catches and reproduce the remainder (double, max 30).

4. **Record the outcome:**

- Fish caught (per person + total) and fish remaining.

Figure 2: Watch this video explaining Maximum Sustainable Yield before you proceed to Round 2.

Round 2 Reflection Prompt: Discuss with your group and respond on a separate paper **before you proceed**.

1. Was it challenging or easy to respect the group quota?
2. Did everyone cooperate, or did someone overfish?
3. How is this scenario more or less

Round 3: Fixed-Effort Harvesting

Scenario

- Now you will limit the **per-person catch** each round (e.g., 2 fish per person if you want a total near ~12 when in a group of 6).
- This time, there is **no secrecy**: the group openly decides how many fish each person will harvest.

Instructions

1. **Collaborate** to set a per-person catch limit (the "fixed effort").
2. **Harvest** exactly that amount for each person.
3. **Reproduction:** Double the remaining fish, up to 30.
4. **Record** fish caught (total) and fish remaining.

Round 3 Reflection Prompt: Discuss with your group and respond on a separate paper **before you proceed**.

1. Did fixed effort make it easier or harder to maintain sustainability over multiple rounds?
2. What if some people took fewer fish to be "extra safe" vs. taking the full allowance?

Final Reflection Prompt: Discuss with your group and respond on a separate paper **before you finish**.

1. Suppose you are a fisheries manager in a large coastal region. Which approach (Fixed Quota, Fixed Effort, or Open Access with certain regulations) would you advocate for and why?

- List at least two advantages and two potential drawbacks of your preferred strategy.

Looking Ahead: Conservation biologists and resource managers often use mathematical models to predict how populations respond under different harvesting regimes. Today's activity was a simplified version, but the lessons about cooperation, communication, and overexploitation apply across many types of shared resources—from forest timber and clean water to global carbon emissions.